PROFESSOR FERRETTO

Física

GRAVITAÇÃO

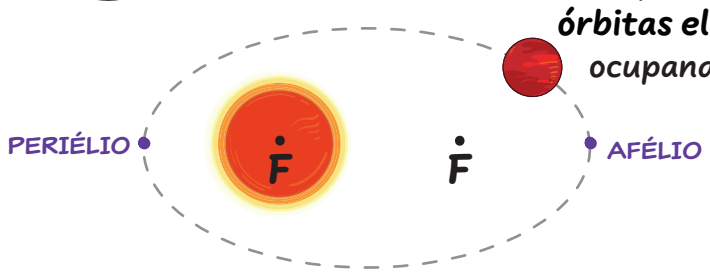
FÍSICA

PROFESSOR COELHO

LEIS DE KEPLER

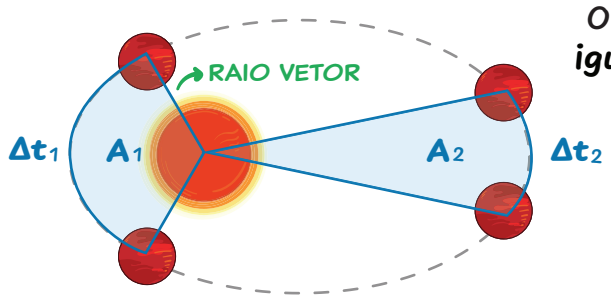
1ª Lei: Lei das Órbitas

Os planetas descrevem **órbitas elípticas**, com o Sol ocupando um dos focos.



2ª Lei: Lei das Áreas

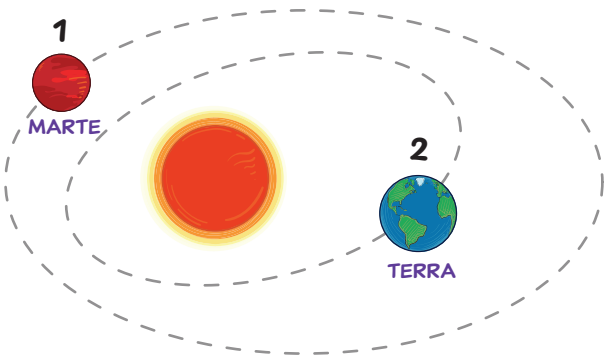
O raio vetor varre **áreas iguais em tempos iguais**



$$\frac{A_1}{\Delta t_1} = \frac{A_2}{\Delta t_2}$$

Quanto **mais perto** do Sol, **maior a velocidade** do planeta.

3ª Lei: Lei dos Períodos

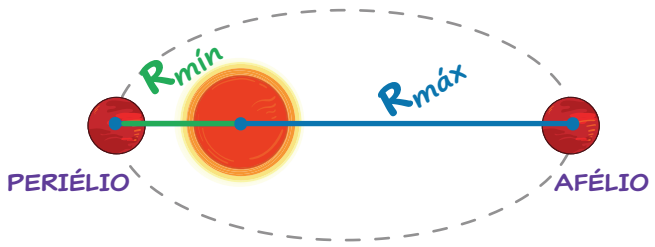


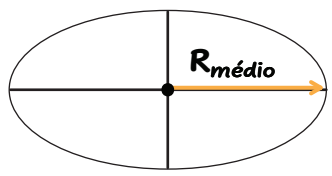
A razão entre o período ao quadrado e o raio médio ao cubo é constante.

$$\frac{(T_1)^2}{(R_1)^3} = \frac{(T_2)^2}{(R_2)^3}$$

RAIO MÉDIO DA ÓRBITA

$$R_{médio} = \frac{R_{máx} + R_{mín}}{2}$$

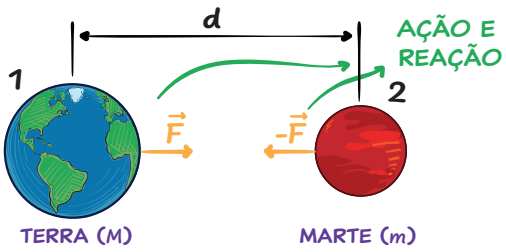




O raio médio é o semi-eixo maior da elipse.

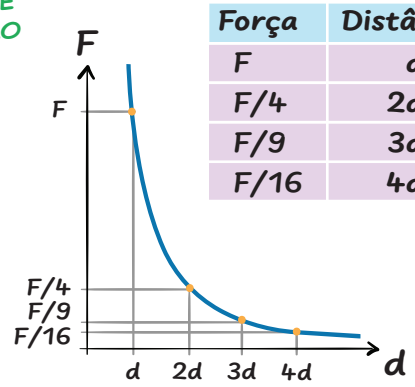
GRAVITAÇÃO UNIVERSAL

Lei da Gravitação Universal



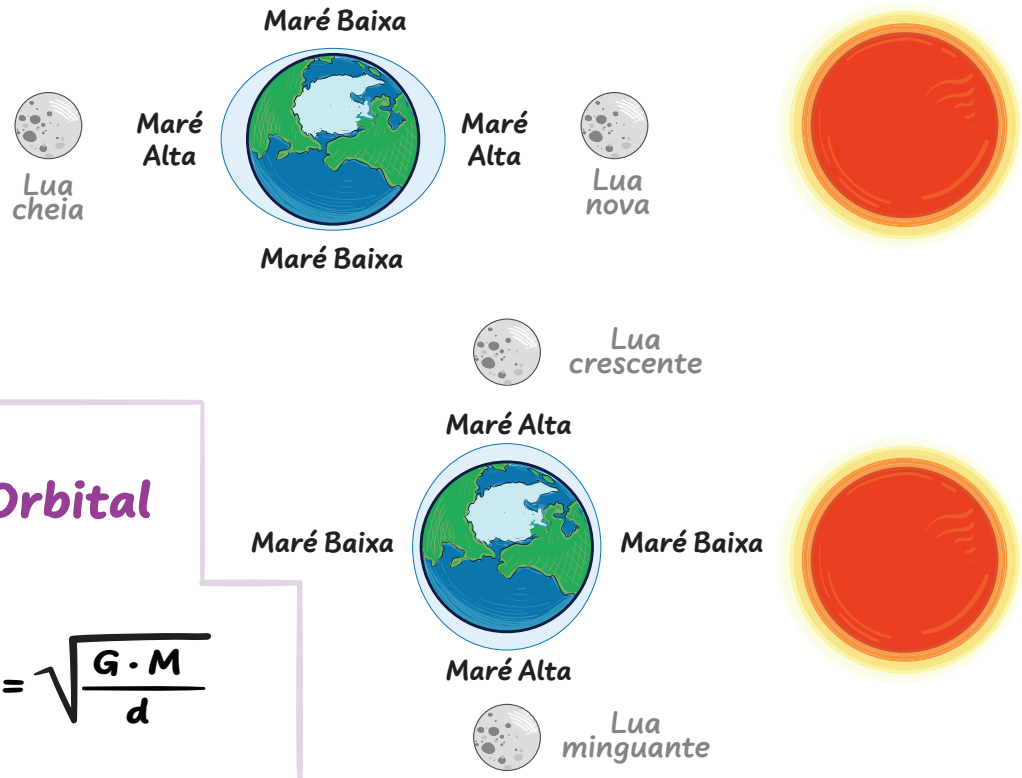
$$F = \frac{G \cdot M \cdot m}{d^2}$$

$$G = 6{,}67 \cdot 10^{-11}\ N \cdot \frac{m^2}{kg^2}$$



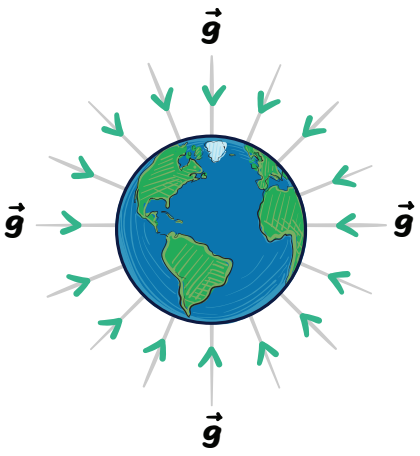
Força	Distância
F	d
F/4	2d
F/9	3d
F/16	4d

Marés



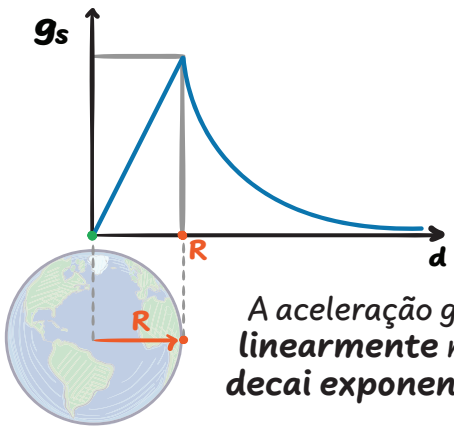
Campo Gravitacional

RADIAL E CONVERGENTE



$$g = \frac{P}{m}$$

$$g = \frac{G \cdot M}{d^2}$$

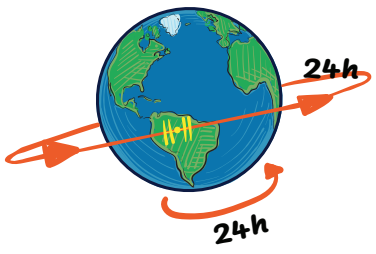


A aceleração gravitacional **cresce linearmente** no interior da Terra e **decai exponencialmente** fora dela.

Velocidade Orbital

$$v = \sqrt{\frac{G \cdot M}{d}}$$

Satélites Geoestacionários



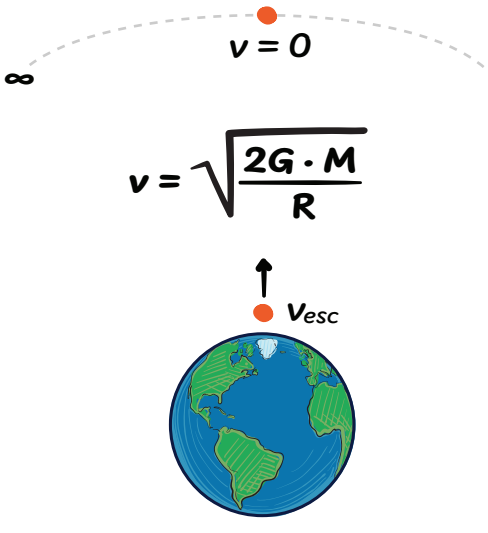
O período da órbita do satélite é igual ao período de rotação da Terra.

Energia Potencial Gravitacional

$$E_{PG} = \frac{-G \cdot M \cdot m}{d}$$

A energia potencial é nula no infinito.

Velocidade de Escape



$$v = \sqrt{\frac{2G \cdot M}{R}}$$